

SYMBIOSIS INSTITUTE
OF TECHNOLOGY (SIT)
Constituent of SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

DelayLens: Agentic Big Data Analytics for Delivery-Delay Intelligence using Hadoop MapReduce, MongoDB and LangGraph

Continuous Assessment 3 (Big Data Case Study Analysis)

Subject: Big Data Analytics

Submitted By:

Arav Saxena
PRN: 23070126018

Mahi Laddha
PRN: 230701260169

Artificial Intelligence and Machine Learning

PROBLEM & OBJECTIVES

54.8% of DataCo orders arrive late

65,752
orders analysed

180,519
order lines

3 years
Jan 2015 to Jan 2018

- Find where and why orders are late
- Test which promise changes would fix it
- Give staff an AI copilot for instant answers

DATASET

DataCo Smart Supply Chain data

180,519
order lines, 53 fields (34 kept)

469,977
web access-log lines

Mendeley Data
public source, three files

3,249
duplicate log lines removed

- Spanish country and state names translated
- Late = actual days above promised days (100% match)

ARCHITECTURE AND TOOLS

Six tools, one pipeline

1 **Clean** — pandas

→

2 **Store** — HDFS 3.5.0, 16 MB blocks

→

3 **Process** — MapReduce on YARN

→

4 **Serve** — MongoDB 7, FastAPI

→

5 **Ask** — LangGraph agent

→

6 **See** — Next.js 16 dashboard

The copilot can send a new MapReduce job back to the cluster

HADOOP IMPLEMENTATION

33 MapReduce jobs, one algorithm

33
jobs on YARN

65,752
orders rolled up from 180,519 lines

2,600x
smaller shuffle with a combiner

6.2x
run time for 64x more data

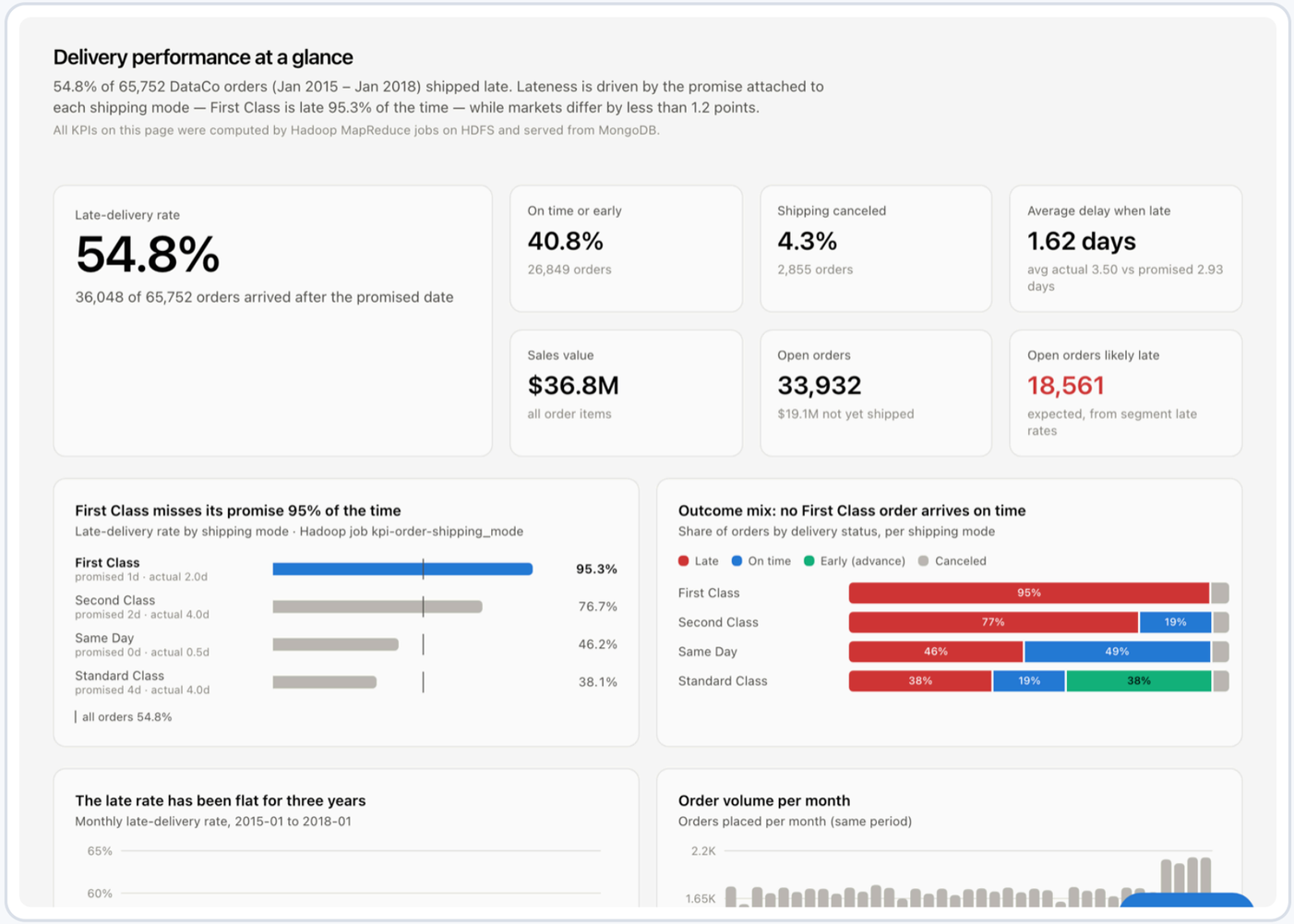
- Job A turns order lines into one record per order
- Native Java is 14% to 28% faster than Streaming
- 25 jobs reuse one map, combine, reduce pattern
- Hadoop, MongoDB and pandas agree: 0 mismatches

DASHBOARD ANALYSIS | OVERVIEW

54.8% late, flat for 3 years

WHAT IT SHOWS

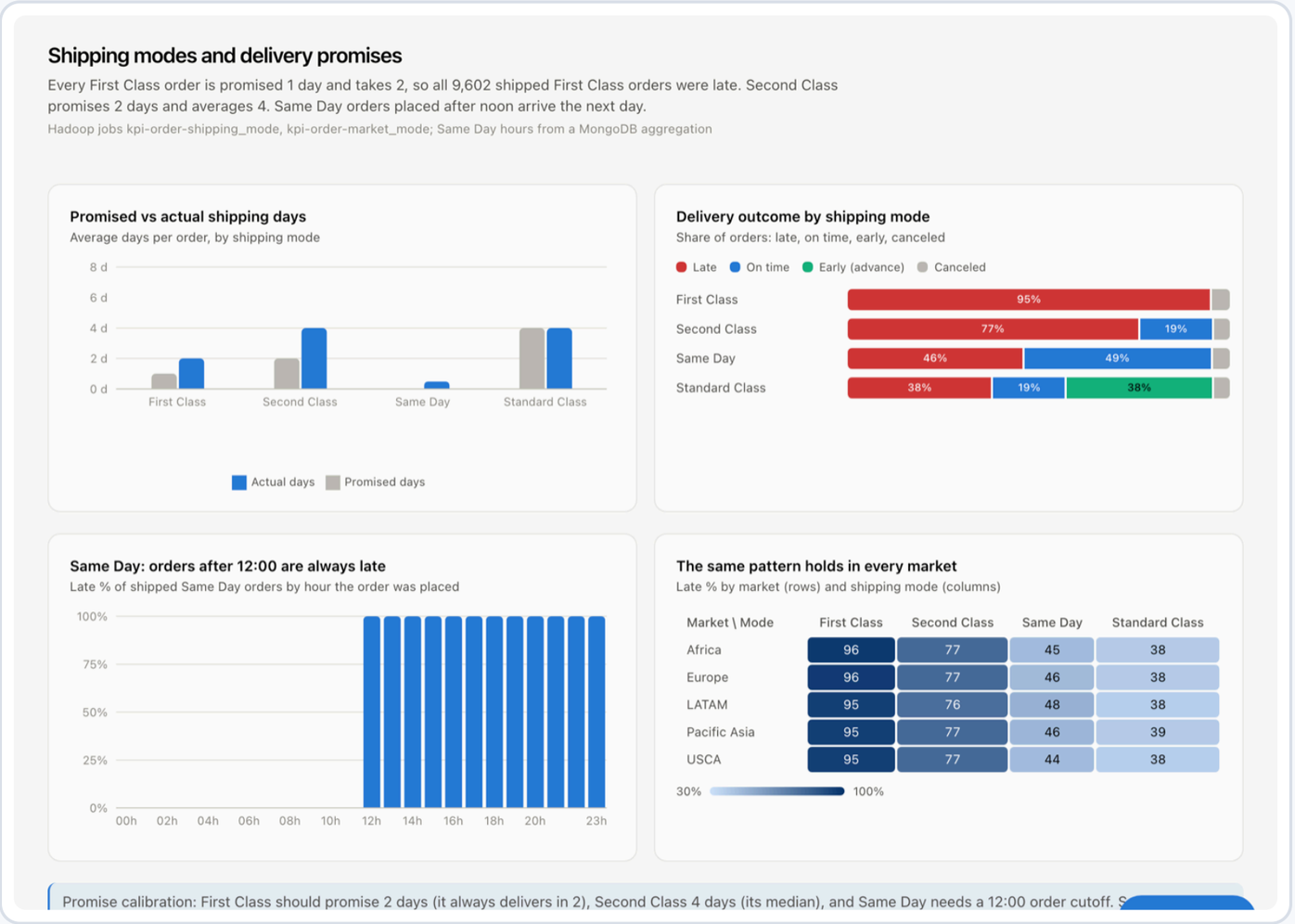
- 36,048 of 65,752 orders late
- Late rate flat across 37 months
- Every market within 2 points of the average
- 33,932 open orders, 18,561 expected late



Shipping mode decides lateness

WHAT IT SHOWS

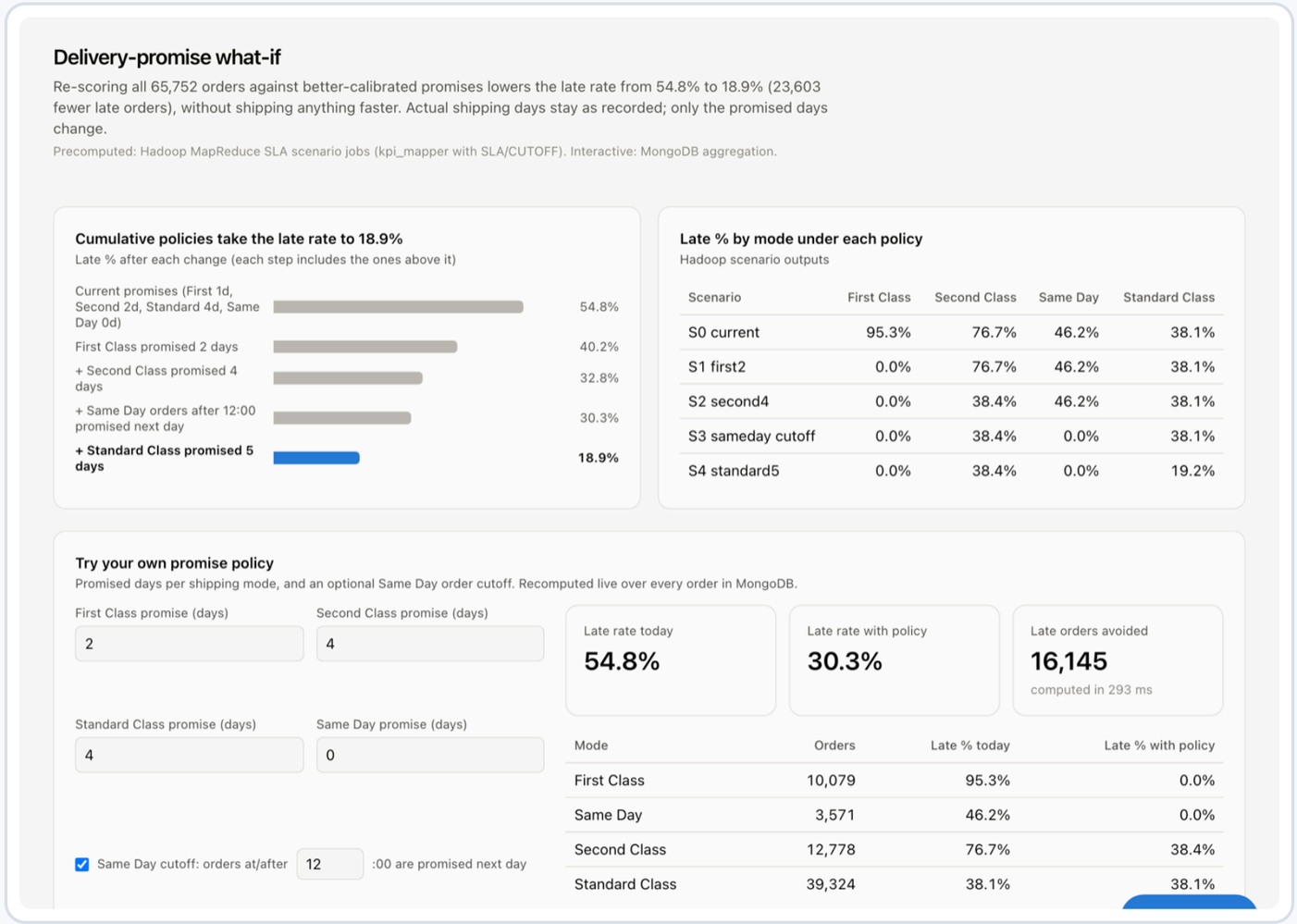
- First Class: promised 1 day, takes 2, 95.3% late
- Same Day orders after 12:00 are always late
- Same pattern in every market
- Standard Class is the most reliable: 38.1%



Better promises: 54.8% to 18.9%

WHAT IT SHOWS

- 23,603 fewer late orders, no faster shipping
- Live simulator: 30.3% in 293 ms
- First Class 2 days, Second Class 4 days, Same Day cut-off 12:00
- 11,674 open orders ($6.58M) at risk



Delivery-promise what-if

Re-scoring all 65,752 orders against better-calibrated promises lowers the late rate from 54.8% to 18.9% (23,603 fewer late orders), without shipping anything faster. Actual shipping days stay as recorded; only the promised days change.

Precomputed: Hadoop MapReduce SLA scenario jobs (kpi_mapper with SLA/CUTOFF). Interactive: MongoDB aggregation.

Cumulative policies take the late rate to 18.9%

Late % after each change (each step includes the ones above it)

Current promises (First 1d, Second 2d, Standard 4d, Same Day 0d)	54.8%
First Class promised 2 days	40.2%
+ Second Class promised 4 days	32.8%
+ Same Day orders after 12:00 promised next day	30.3%
+ Standard Class promised 5 days	**18.9%**

Late % by mode under each policy

Hadoop scenario outputs

Scenario	First Class	Second Class	Same Day	Standard Class
S0 current	95.3%	76.7%	46.2%	38.1%
S1 first2	0.0%	76.7%	46.2%	38.1%
S2 second4	0.0%	38.4%	46.2%	38.1%
S3 sameday cutoff	0.0%	38.4%	0.0%	38.1%
S4 standard5	0.0%	38.4%	0.0%	19.2%

Try your own promise policy

Promised days per shipping mode, and an optional Same Day order cutoff. Recomputed live over every order in MongoDB.

First Class promise (days): 2 | Second Class promise (days): 4

Standard Class promise (days): 4 | Same Day promise (days): 0

☑ Same Day cutoff: orders at/after 12 :00 are promised next day

Late rate today: 54.8%

Late rate with policy: 30.3%

Late orders avoided: 16,145 computed in 293 ms

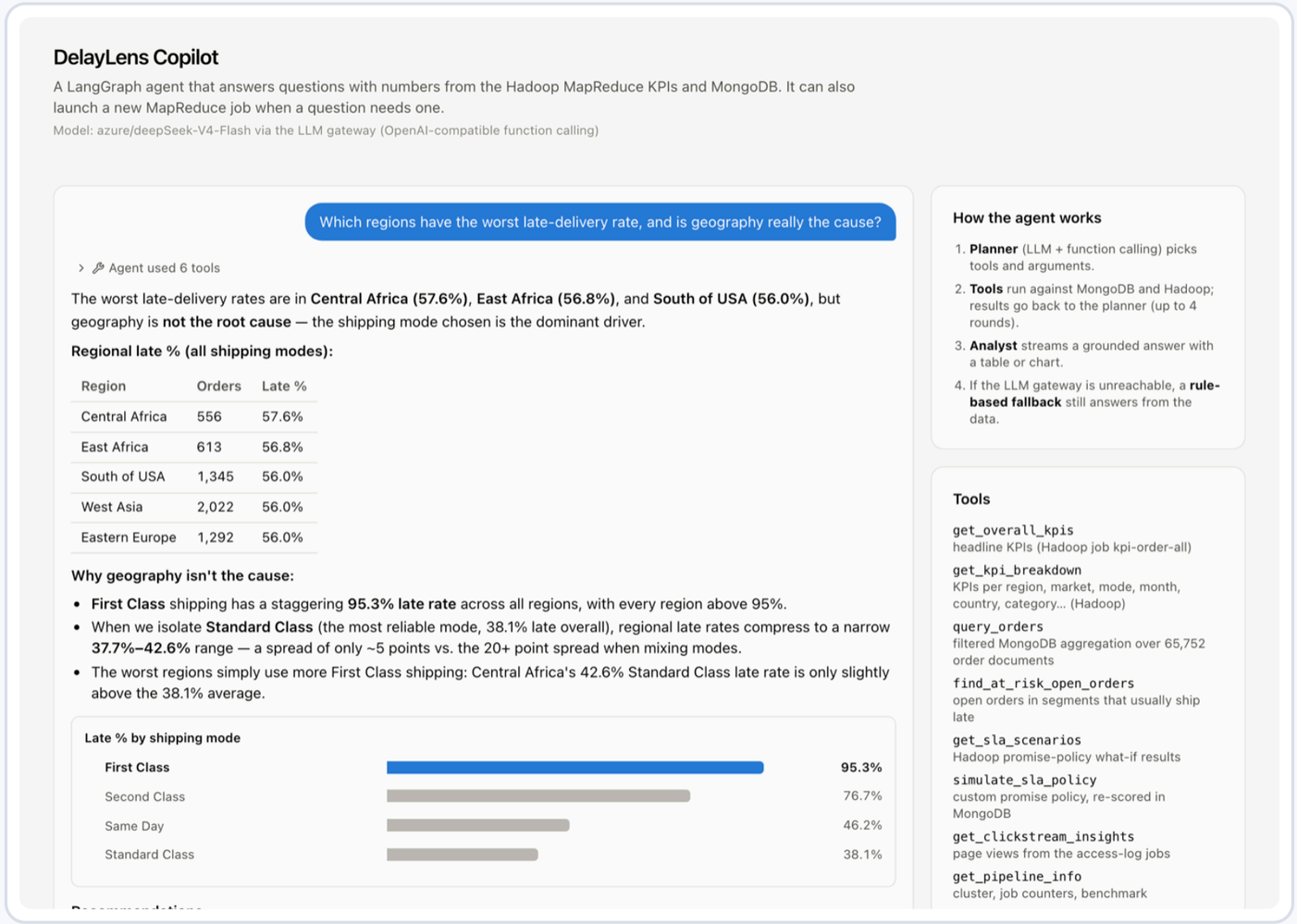
Mode	Orders	Late % today	Late % with policy
First Class	10,079	95.3%	0.0%
Same Day	3,571	46.2%	0.0%
Second Class	12,778	76.7%	38.4%
Standard Class	39,324	38.1%	38.1%

DASHBOARD ANALYSIS | COPILOT

The copilot explains the cause

WHAT IT SHOWS

- Asked: is geography the cause?
- Agent used 6 tools and drew a chart
- Answer: shipping mode, not region
- 8 of 8 test questions correct, 4 to 7 s each



CONCLUSION

Fix the promise, not the map

Findings

- Delays follow shipping-mode promises
- Hadoop + MongoDB + copilot work end to end
- Copilot: 8 of 8 answers correct

Recommendations

- Promise First Class in 2 days
- Promise Second Class in 4 days
- Contact the 11,674 at-risk open orders

Future work: multi-node cluster and live data feeds · Thank you